

多元微分学与多元积分学

【题 1】 若 $f(x, y)$ 是在 $\mathbb{R}^2 \setminus \{0\}$ 上连续的 n 次齐次函数 (注: n 次齐次函数是指满足 $f(t\mathbf{x}) = t^n f(\mathbf{x})$ 的函数), 证明: $|f(x, y)| \leq C(x^2 + y^2)^{\frac{n}{2}}$, 其中 C 为正常数

证明: 考虑集合 $S = \{(x, y) \mid x^2 + y^2 = 1\}$, 则 S 为有界闭集,

又 f 为连续函数, $\therefore |f|$ 也是连续函数, $\therefore \exists C > 0$, $(x, y) \in S$ 时, $|f(x, y)| \leq C$,

对 $\forall (x, y) \in \mathbb{R}^2 \setminus \{0\}$, 令 $r = \sqrt{x^2 + y^2}$, 则 $f(x, y) = f(r \cdot \frac{x}{r}, r \cdot \frac{y}{r}) = r^n f(\frac{x}{r}, \frac{y}{r})$,

又 $\because (\frac{x}{r}, \frac{y}{r}) \in S$, $\therefore \left|f(\frac{x}{r}, \frac{y}{r})\right| \leq C$,

$\therefore |f(x, y)| \leq C(x^2 + y^2)^{\frac{n}{2}}$.

【题 2】 如果 $F(x, y)$ 在矩形 $D = \{(x, y) \in \mathbb{R}^2 \mid |x - x_0| < a, |y - y_0| < b\}$ 上连续, $F(x_0, y_0) = 0$, 且 $F(x, y)$ 对每一个 $x \in (x_0 - a, x_0 + a)$ 关于 y 严格单调, 求证: 在 (x_0, y_0) 某个邻域内, 由 $F(x, y) = 0$ 可确定唯一的函数 $y = f(x)$, 且 $f(x)$ 在 x_0 的一个邻域里连续

证明: 不妨设关于 y 严格单调递增. 取 $(x_0, y_1), (x_0, y_2) \in D$, 且 $y_1 > y_0 > y_2$,

则 $F(x_0, y_1) > 0, F(x_0, y_2) < 0$. 取 $0 < \alpha < a$ 使 $S = [x_0 - \alpha, x_0 + \alpha] \times [y_2, y_1] \subset D$,

因为 F 在闭矩形 S 上一致连续, 令 $\varepsilon_1 = \frac{\min\{|F(x_0, y_1)|, |F(x_0, y_2)|\}}{2}$,

可取 $\delta_0 > 0$, 使 $u, v \in S$ 且 $\|u - v\| < \delta_0$ 时, $|F(u) - F(v)| < \varepsilon_1$,

令 $\delta = \min\{\delta_0, \alpha\}$, 对每个 $x \in (x_0 - \delta, x_0 + \delta)$, 有 $F(x, y_1) > 0, F(x, y_2) < 0$,

故由连续性和严格单调性, 存在唯一的 $f(x) \in (y_2, y_1)$ 使 $F(x, f(x)) = 0$, 且 $f(x_0) = y_0$,

下证 f 连续, 任取 $x^* \in (x_0 - \delta, x_0 + \delta)$ 及 $\varepsilon > 0$,

取 $0 < \rho < \min\{\varepsilon, f(x^*) - y_2, y_1 - f(x^*)\}$,

则由严格单调性, $F(x^*, f(x^*) - \rho) < 0 < F(x^*, f(x^*) + \rho)$,

由 F 的连续性, 存在 $\eta_0 > 0$, 使 $|x - x^*| < \eta_0$ 时上述两端仍分别为负和正,

令 $\eta = \min\{\eta_0, \delta - |x^* - x_0|\} > 0$, 当 $|x - x^*| < \eta$ 时,

唯一零点 $f(x)$ 位于 $(f(x^*) - \rho, f(x^*) + \rho)$ 内, 故 $|f(x) - f(x^*)| < \rho < \varepsilon$,

所以 f 在 $(x_0 - \delta, x_0 + \delta)$ 上连续, 命题得证.

【题 3】 $D = \left\{ u \in \mathbb{R}^n \mid \|u\| \leq \frac{1}{2} \right\}$, $f: D \rightarrow \mathbb{R}$ 为 C^1 映射,
 $\|\nabla f(0)\| = 1$, 且 $\|\nabla f(u) - \nabla f(v)\| \leq \|u - v\|$,
 证明: 存在唯一的点 ξ , 使得 $f(\xi) = \max_D f$

证明: $\because D$ 为有界闭集, f 为连续映射, 从而 f 在 D 上能取到最大值,

当 $u \in \text{int}D$ 时, $\|\nabla f(u) - \nabla f(0)\| \leq \|u\| < \frac{1}{2} \Rightarrow \frac{1}{2} < \|\nabla f(u)\| < \frac{3}{2}$,

特别地, $\|\nabla f(u)\| \neq 0$, 从而 f 的最大值只能在 ∂D 上取到,

令 $F(u) = f(u) - \frac{1}{2}\|u\|^2$, $\therefore \nabla F(u) = \nabla f(u) - u$, 同理 F 可在 D 上取到最大值,

当 $u \in \text{int}D$ 时, $\|\nabla F(u)\| = \|\nabla f(u) - u\| \geq \|\nabla f(u)\| - \|u\| > 0$,

$\therefore \nabla F(u) \neq 0$, 即 F 的最大值也只能在 ∂D 上取到,

又 $\max_D F = \max_{\partial D} F = \max_{\partial D} f - \frac{1}{8} = \max_D f - \frac{1}{8}$,

$\therefore F$ 与 f 有完全相同的最大值点, 因此只需要证明 F 的最大值点唯一即可,

对任意 $u, v \in D$, 由条件与 *Cauchy* 不等式,

$$\begin{aligned} & (\nabla F(u) - \nabla F(v)) \cdot (u - v) \\ &= (\nabla f(u) - \nabla f(v)) \cdot (u - v) - \|u - v\|^2 \\ &\leq \|\nabla f(u) - \nabla f(v)\| \|u - v\| - \|u - v\|^2 \leq 0, \end{aligned}$$

令 $w(t) = (1 - t)u + tv$, $\psi(t) = F(w(t))$, 若 $0 \leq s < t \leq 1$,

则 $(\nabla F(w(t)) - \nabla F(w(s))) \cdot (w(t) - w(s)) \leq 0$,

而 $w(t) - w(s) = (t - s)(v - u)$, 故 $\psi'(t) \leq \psi'(s)$,

因此 ψ' 单调不增, ψ 为凹函数, 从而 F 在 D 上为凹函数,

假设 u^0, v^0 是 F 的两个不同最大值点, 记 $F(u^0) = F(v^0) = M$,

对任意 $t \in (0, 1)$, 由凹性与 M 的最大性,

$$M \geq F((1 - t)u^0 + tv^0) \geq (1 - t)F(u^0) + tF(v^0) = M,$$

故线段开段上每一点都是 F 的最大值点,

由于闭球 D 严格凸, $(1 - t)u^0 + tv^0 \in \text{int } D$,

由 *Fermat* 必要条件可知该点满足 $\nabla F = 0$, 与前面已证的 $\nabla F \neq 0$ 矛盾,

从而 F 只能有唯一的最大值点, 即存在唯一的点 ξ , 使得 $f(\xi) = \max_D f$, 得证.

【题 4】 二元函数 $f(x, y)$, 定义域为 $I = [a, b] \times [c, d]$, f 关于每个变量单调递增, 证明: $f \in R(I)$

证明: 取等距分划 $a = x_0 < \cdots < x_n = b$, $c = y_0 < \cdots < y_n = d$, 并令

$$R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j], \quad \Delta x = \frac{b-a}{n}, \quad \Delta y = \frac{d-c}{n}.$$

由单调性, f 在 R_{ij} 上的振幅为 $\omega_{ij} = f(x_i, y_j) - f(x_{i-1}, y_{j-1})$

$$= [f(x_i, y_j) - f(x_{i-1}, y_j)] + [f(x_{i-1}, y_j) - f(x_{i-1}, y_{j-1})],$$

因此沿 i, j 分别求和, 可得 $S(f, P_n) - s(f, P_n) = \sum_{i,j} \omega_{ij} \Delta x \Delta y$

$$\begin{aligned} &= \Delta x \Delta y \left\{ \sum_{j=1}^n [f(b, y_j) - f(a, y_j)] + \sum_{i=1}^n [f(x_{i-1}, d) - f(x_{i-1}, c)] \right\} \\ &\leq \frac{2(b-a)(d-c)}{n} [f(b, d) - f(a, c)] \rightarrow 0. \end{aligned}$$

故对任意 $\varepsilon > 0$, 可取充分大的 n , 使 $S(f, P_n) - s(f, P_n) < \varepsilon$, 从而 $f \in R(I)$, 得证.

【题 5】 $\gamma(t) : [a, b] \rightarrow \mathbb{R}^2, \gamma(t) = (x(t), y(t))$, 且 $x(t) \in C^\alpha[a, b], y(t) \in C^\beta[a, b], 0 < \alpha, \beta < 1, \alpha + \beta > 1$, 证明: $\gamma([a, b])$ 是 $\mathbb{R}^2$ 中零面积集. (注: $x(t) \in C^\alpha[a, b]$ 指的是 $|x(t_1) - x(t_2)| \leq C_x |t_1 - t_2|^\alpha$)

证明: 将 $[a, b]$ n 等分, $a = t_0 < t_1 < \cdots < t_n = b, \Delta t_i = \frac{b-a}{n}$,

设 $x(t)$ 在 $[t_{i-1}, t_i]$ 上最大值, 最小值分别为 α_i, β_i ,

$y(t)$ 在 $[t_{i-1}, t_i]$ 上最大值, 最小值分别为 μ_i, ν_i ,

令 $I_i = [\beta_i, \alpha_i] \times [\nu_i, \mu_i]$, 则 $\gamma([a, b]) \subset \bigcup_{i=1}^n I_i$, 取 $C = \max\{C_x, C_y\}$,

$$\because \alpha_i - \beta_i \leq C |t_i - t_{i-1}|^\alpha = \frac{C(b-a)^\alpha}{n^\alpha},$$

$$\mu_i - \nu_i \leq C |t_i - t_{i-1}|^\beta = \frac{C(b-a)^\beta}{n^\beta},$$

$$\therefore \sum_{i=1}^n \sigma(I_i) = \sum_{i=1}^n \frac{C^2(b-a)^{\alpha+\beta}}{n^{\alpha+\beta}} = \frac{C^2(b-a)^{\alpha+\beta}}{n^{\alpha+\beta-1}},$$

$$\because \alpha + \beta > 1, \quad \therefore \lim_{n \rightarrow \infty} \frac{C^2(b-a)^{\alpha+\beta}}{n^{\alpha+\beta-1}} = 0,$$

$$\therefore \forall \varepsilon > 0, \exists N, \text{ s.t. } \forall n \geq N, \gamma([a, b]) \subset \bigcup_{i=1}^n I_i, \text{ 且 } \sum_{i=1}^n \sigma(I_i) < \varepsilon,$$

从而 $\gamma([a, b])$ 是 $\mathbb{R}^2$ 中零面积集, 得证.

【题 6】 设 $f \in C^2(\overline{B_1(0)})$, 在 $B_1(0)$ 上 C^2 , 设 f 在 $\partial B_1(0)$ 上最小值为 a , $f(0) = b < a$, 考虑映射 $\phi : B_1(0) \rightarrow \mathbb{R}^n, \phi(x) = \nabla f(x), \Sigma = \{x \in B_1(0) | \nabla^2 f(x) \text{ 半正定} \}$, 证明: $\phi(\Sigma)$ 中包含一个半径为 $a - b$ 的开球

证明: $\because f(0) = b < a = \min_{\partial B_1(0)} f$, 且 $\overline{B_1(0)}$ 是有界闭集, $f \in C^2(\overline{B_1(0)})$,

$\therefore f$ 在 $\overline{B_1(0)}$ 上有最小值, 且最小值点在 $B_1(0)$ 取到,

即 $\exists x^* \in B_1(0)$, s.t. $\nabla f(x^*) = 0$, 且 $\nabla^2 f(x^*) \geq 0$, $\therefore x^* \in \Sigma$,

$\therefore 0 = \phi(x^*)$, $\therefore 0 \in \phi(\Sigma)$,

对 $\forall v \in B_{a-b}(0)$, 考虑函数 $g(x) = f(x) - v \cdot x$, 则 $g \in C^0(\overline{B_1(0)})$, 且 $g(0) = f(0) = b$,

$x \in \partial B_1(0)$ 时, $g(x) \geq a - v \cdot x \geq a - \|v\| \|x\| > a - (a - b) = b$,

从而 $g(x)$ 在 $\overline{B_1(0)}$ 上的最小值在 $B_1(0)$ 取到,

即 $\exists y^0 \in B_1(0)$, s.t. $\nabla g(y^0) = 0$ 且 $\nabla^2 g(y^0) \geq 0$,

又 $\nabla g(y^0) = \nabla f(y^0) - v$, $\nabla^2 g(y^0) = \nabla^2 f(y^0) \geq 0$,

$\therefore y^0 \in \Sigma \quad \therefore v = \nabla f(y^0) \in \phi(\Sigma)$,

从而 $B_{a-b}(0) \subset \phi(\Sigma)$, 证毕.

【题 7】 设 $D \subset \mathbb{R}^2$ 为平面区域, $u \in C^2(D)$, 满足 $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, 证明: 对任何 $\overline{B_r(a)} \subset D$, 都有 $\int_{\partial B_r(a)} \frac{\partial u}{\partial \vec{n}} ds = 0$, 其中 $\vec{n}$ 为 $\partial B_r(a)$ 的单位外法向量, $\frac{\partial u}{\partial \vec{n}}$ 为方向导数.

证明: 设 $a = (a_1, a_2)$, 则 $\partial B_r(a): \begin{cases} x = a_1 + r \cos \theta \\ y = a_2 + r \sin \theta \end{cases} \quad \theta \in [0, 2\pi]$,

$$\therefore \vec{n} = (\cos \theta, \sin \theta), \quad ds = r d\theta,$$

$$\text{又 } u \in C^2(D), \quad \therefore \frac{\partial u}{\partial \vec{n}} = \nabla u \cdot \vec{n} = \frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta,$$

$$\begin{aligned} \therefore \int_{\partial B_r(a)} \frac{\partial u}{\partial \vec{n}} ds &= \int_0^{2\pi} \frac{\partial u}{\partial x} (a_1 + r \cos \theta, a_2 + r \sin \theta) r \cos \theta d\theta \\ &+ \int_0^{2\pi} \frac{\partial u}{\partial y} (a_1 + r \cos \theta, a_2 + r \sin \theta) r \sin \theta d\theta \\ &= \int_0^{2\pi} \left[\frac{\partial u}{\partial x} \cdot (r \cos \theta) + \frac{\partial u}{\partial y} \cdot (r \sin \theta) \right] d\theta = \int_0^{2\pi} \left[\frac{\partial u}{\partial x} \cdot y'(\theta) - \frac{\partial u}{\partial y} \cdot x'(\theta) \right] d\theta \\ &= \oint_{\partial B_r(a)} \frac{\partial u}{\partial x} dy - \frac{\partial u}{\partial y} dx, \end{aligned}$$

$$\text{设 } P(x, y) = -\frac{\partial u}{\partial y}, \quad Q(x, y) = \frac{\partial u}{\partial x}, \quad \therefore P, Q \in C^1(D), \quad \frac{\partial Q}{\partial x} = \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial P}{\partial y} = -\frac{\partial^2 u}{\partial y^2},$$

$$\text{由 Green 公式, } \oint_{\partial B_r(a)} \frac{\partial u}{\partial x} dy - \frac{\partial u}{\partial y} dx = \iint_{B_r(a)} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) dx dy = 0,$$

$$\therefore \int_{\partial B_r(a)} \frac{\partial u}{\partial \vec{n}} ds = 0, \text{ 得证.}$$

【题 8】 Ω 为有界 C^1 区域, $u \in C^2(\bar{\Omega})$, $\Delta u = u$, 且 $u|_{\partial\Omega} \equiv 0$, 证明: $u \equiv 0$

微分学视角: 假设 u 不恒为 0. 不妨设 u 的最大值大于 0, 否则对 $-u$ 作同样论证. 由边界条件, 存在 $x^0 \in \text{int}\Omega$, s.t. $u(x^0) > 0$,

$$\text{且 } u(x^0) = \max_{\Omega} u, \text{ 则 } \nabla u(x^0) = 0, \text{ 且 } \nabla^2 u(x^0) \leq 0,$$

$$\text{由于 } \Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \text{tr}(\nabla^2 u),$$

$$\therefore \Delta u(x^0) \leq 0, \text{ 与 } \Delta u(x^0) = u(x^0) \text{ 矛盾! 从而 } u \equiv 0, \text{ 得证.}$$

$$\text{积分学视角: 考虑积分 } \int_{\Omega} u^2 dx dy, \text{ 则 } \int_{\Omega} u^2 dx dy = \int_{\Omega} u \Delta u dx dy,$$

$$\therefore \text{div}(u \nabla u) = \text{div}\left(u \frac{\partial u}{\partial x}, u \frac{\partial u}{\partial y}\right) = \frac{\partial(u \frac{\partial u}{\partial x})}{\partial x} + \frac{\partial(u \frac{\partial u}{\partial y})}{\partial y}$$

$$= u \frac{\partial^2 u}{\partial x^2} + \left(\frac{\partial u}{\partial x}\right)^2 + u \frac{\partial^2 u}{\partial y^2} + \left(\frac{\partial u}{\partial y}\right)^2 = u \Delta u + \|\nabla u\|^2,$$

$$\text{由 Gauss 公式, } \int_{\Omega} u \Delta u dx dy = \int_{\Omega} \operatorname{div}(u \nabla u) dx dy - \int_{\Omega} \|\nabla u\|^2 dx dy$$

$$= \int_{\partial \Omega} u \nabla u \cdot \vec{n} ds - \int_{\Omega} \|\nabla u\|^2 dx dy,$$

$$= - \int_{\Omega} \|\nabla u\|^2 dx dy \leq 0, \text{ 而 } \int_{\Omega} u^2 dx dy \geq 0,$$

$$\text{从而 } \int_{\Omega} u^2 dx dy = 0 \Rightarrow u \equiv 0, \text{ 得证.}$$

【题 9】计算 $\oint_{\Gamma} \frac{x dy - y dx}{ax^2 + 2cxy + by^2}$, 其中 $\Gamma: x^2 + y^2 = 1$, 方向为逆时针且 $ab > c^2$

证明: 分母 $ax^2 + 2cxy + by^2 = (x, y)M \begin{pmatrix} x \\ y \end{pmatrix}$,

其中 $M = \begin{pmatrix} a & c \\ c & b \end{pmatrix}$ 为实对称矩阵, 且 $\det M = ab - c^2 > 0$,

可选 $P \in SO(2)$, 使 $P^T M P = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$,

令 $\begin{pmatrix} x \\ y \end{pmatrix} = P \begin{pmatrix} u \\ v \end{pmatrix}$, 则单位圆及其逆时针方向保持不变, 且 $x dy - y dx = u dv - v du$,

由 $ab > c^2$ 可知 a, b 同号, 两个特征值也与 a 同号,

令 $s = \operatorname{sgn}(a)$, 则可写 $\lambda_i = s\mu_i$, 其中 $\mu_i > 0$, 且 $\mu_1 \mu_2 = \lambda_1 \lambda_2 = ab - c^2$,

取 $u = \cos \theta, v = \sin \theta$, 则 $\oint_{\Gamma} \frac{x dy - y dx}{ax^2 + 2cxy + by^2}$

$$= s \int_0^{2\pi} \frac{d\theta}{\mu_1 \cos^2 \theta + \mu_2 \sin^2 \theta} = 4s \int_0^{\frac{\pi}{2}} \frac{d\theta}{\mu_1 \cos^2 \theta + \mu_2 \sin^2 \theta},$$

$$\text{令 } t = \tan \theta, \text{ 则 } 4s \int_0^{\frac{\pi}{2}} \frac{d\theta}{\mu_1 \cos^2 \theta + \mu_2 \sin^2 \theta} = 4s \int_0^{+\infty} \frac{dt}{\mu_1 + \mu_2 t^2} = \frac{2\pi s}{\sqrt{\mu_1 \mu_2}},$$

$$\text{故: } \boxed{\oint_{\Gamma} \frac{x dy - y dx}{ax^2 + 2cxy + by^2} = \frac{2\pi \operatorname{sgn}(a)}{\sqrt{ab - c^2}}}.$$

【题 10】 $\Omega \subset \mathbb{R}^3$ 为有 C^1 边界的有界区域, $u \in C^2(\bar{\Omega}, \mathbb{R})$, $\Delta u = 0$, 证明: 对任意内点 $p \in \Omega$, 有: $u(p) = \frac{1}{4\pi} \int_{\partial\Omega} \left(\frac{\mathbf{r} \cdot \mathbf{n}}{r^3} u + \frac{1}{r} \frac{\partial u}{\partial \mathbf{n}} \right) d\sigma$, 其中 $p = (x_0, y_0, z_0)$, $\mathbf{r} = (x - x_0, y - y_0, z - z_0)$, $\mathbf{n}$ 为 $\partial\Omega$ 单位外法向量

证明: 固定 $\varepsilon_0 > 0$ 使 $\overline{B_{\varepsilon_0}(p)} \subset \Omega$, 并取 $0 < \varepsilon < \varepsilon_0$. 因为 $\frac{\partial u}{\partial \mathbf{n}} = \nabla u \cdot \mathbf{n}$, 令 $\mathbf{X} = \frac{\mathbf{r}}{r^3} u + \frac{1}{r} \nabla u$, 考虑 $\Omega \setminus B_\varepsilon(p)$,

$$\text{记 } I_1 = \int_{\partial\Omega} \left(\frac{\mathbf{r} \cdot \mathbf{n}}{r^3} u + \frac{1}{r} \frac{\partial u}{\partial \mathbf{n}} \right) d\sigma, \quad I_2 = \int_{\partial B_\varepsilon(p)} \left(\frac{\mathbf{r} \cdot \mathbf{n}}{r^3} u + \frac{1}{r} \frac{\partial u}{\partial \mathbf{n}} \right) d\sigma,$$

则由 Gauss 公式, $I_1 + I_2 = \int_{\Omega \setminus B_\varepsilon(p)} \operatorname{div} \mathbf{X} \, dx dy dz$, 记 $\mathbf{X} = (P, Q, R)$,

$$\begin{aligned} \therefore \frac{\partial P}{\partial x} &= \frac{1}{r} \frac{\partial^2 u}{\partial x^2} + \left(-\frac{1}{r^2} \right) \frac{x - x_0}{r} \frac{\partial u}{\partial x} + \frac{x - x_0}{r^3} \frac{\partial u}{\partial x} + u \cdot \frac{r^2 - 3(x - x_0)^2}{r^5} \\ &= \frac{1}{r} \frac{\partial^2 u}{\partial x^2} + u \cdot \frac{r^2 - 3(x - x_0)^2}{r^5}, \end{aligned}$$

$$\text{同理 } \frac{\partial Q}{\partial y} = \frac{1}{r} \frac{\partial^2 u}{\partial y^2} + u \cdot \frac{r^2 - 3(y - y_0)^2}{r^5}, \quad \frac{\partial R}{\partial z} = \frac{1}{r} \frac{\partial^2 u}{\partial z^2} + u \cdot \frac{r^2 - 3(z - z_0)^2}{r^5},$$

$$\therefore \operatorname{div} \mathbf{X} = \frac{1}{r} \Delta u + u \cdot \frac{3r^2 - 3r^2}{r^5} = 0, \quad \therefore I_1 + I_2 = 0,$$

在 $\partial B_\varepsilon(p)$ 上, 诱导定向的 $\mathbf{n} = -\frac{\mathbf{r}}{r}$, 且 $r = \varepsilon$,

$$\therefore I_2 = - \int_{\partial B_\varepsilon(p)} \left(\frac{u}{\varepsilon^2} + \frac{\nabla u}{\varepsilon} \cdot \frac{\mathbf{r}}{r} \right) d\sigma,$$

令 $C = \max_{\overline{B_{\varepsilon_0}(p)}} \|\nabla u\| < \infty$. 对每个 $0 < \varepsilon < \varepsilon_0$, 有

$$\left| \int_{\partial B_\varepsilon(p)} \frac{\nabla u}{\varepsilon} \cdot \frac{\mathbf{r}}{r} d\sigma \right| \leq \frac{C}{\varepsilon} 4\pi \varepsilon^2 = 4\pi C \varepsilon \rightarrow 0,$$

又由 u 在 p 处连续, 有 $\sup_{|x-p|=\varepsilon} |u(x) - u(p)| \rightarrow 0$,

$$\text{从而 } \left| \frac{1}{\varepsilon^2} \int_{\partial B_\varepsilon(p)} u d\sigma - 4\pi u(p) \right| \leq 4\pi \sup_{|x-p|=\varepsilon} |u(x) - u(p)| \rightarrow 0,$$

$\therefore \lim_{\varepsilon \rightarrow 0} I_2 = -4\pi u(p)$, 又 $I_1 + I_2 = 0$ 对每个 $0 < \varepsilon < \varepsilon_0$ 成立,

$$\therefore I_1 = 4\pi u(p), \text{ 即 } u(p) = \frac{1}{4\pi} \int_{\partial\Omega} \left(\frac{\mathbf{r} \cdot \mathbf{n}}{r^3} u + \frac{1}{r} \frac{\partial u}{\partial \mathbf{n}} \right) d\sigma.$$